# Investigation on the usefulness of investment on a Master's in Data Science

Exploring the financial viability of pursuing a Master's degree in Data Science by analyzing the relationship between educational attainment, salary levels, job roles, experience, and geographical factors.

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Bachelor's in Statistics

Statistical Methods - IV

Kolkata, April 2025

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*Project*

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# Declaration of Authorship

Has undersigned, hereby it his declared that this work entitled "Investigation on the usefulness of investment on a Master's in Data Science" is the original work and that it has not previously in its entirety or in part been submitted at any university or higher education institution for the award of any degree, diploma, or other qualifications. It is also hereby declared that to the best of the knowledge, this work contains no material previously published or written by another person, except where due reference, acknowledgement, and citation is made.

*Kolkata, April 2025*

Abhik Rana

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# Abstract

This study explores the financial viability of pursuing a Master's degree in Data Science by analyzing the relationship between educational attainment, salary levels, job roles, experience, and geographical factors. Through comprehensive data exploration and visual analytics, it examines how advanced education influences earning potential and career trajectory in the data science domain. The findings suggest that while higher education is often associated with increased salaries, the benefits vary significantly depending on location, industry, and prior experience. This analysis offers practical insights for individuals considering graduate education as a pathway to professional advancement in data science.

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# 1

# Introduction

There are one set of people who wants to pursue higher education degrees due to their passion and interest. For these people, it makes sense to get enrol in the relevant university courses and pursue their passion. On the other hand, there is another set of people, who wants to obtain these degrees only to get a specific job title, a specific job role, or a position that get them more money. For this set, university degrees are not the only option, there are many alternatives which can also result in the same outcomes.

The most obvious example is in the field of Data Science and Analytics. In recent years, university degrees such as "Masters in Data Science" or "Masters in Analytics" are sought as one of the must-haves to enter into this field. It is not astonishing that Data Scientist is one of the fastest-growing job titles across the globe and the demand for skilled data scientists is increasing. This has given the universities an option to attract students and make immense money. Several universities have started dedicated degree courses specialized in data science and analytics. Those want to become a data scientist or to switch from another profession to data science profession are now strongly considering these university degrees as the only pathway.

But these university courses are not easy to get in and affordable for everyone. These degrees don't come for free, tuition fees can be exorbitant and can range anything from USD 30,000 to USD 100,000. And that doesn't include the actual cost of living. Many consider applying for student loans but they add a huge lump sum to the existing mountain of debts. A common myth is that the earning potential for those with postgraduate qualifications is higher but of course, there is no guarantee that one will get a stable job at the end of it. Additionally, Pursuing a university's higher degree takes anything from one to three years, depending on different factors. This can seem like a long time, especially when the fellow peers are getting started on their careers, while one is still studying.

The question of interest here is - **does one need to get that expensive higher education degree, do they create a difference from those who do not have university degrees?**. Some resources online also suggest that one can get the depth of knowledge, variety of skills and learn something new. But again, **is it possible to get the same skills, same profile, or even better compensation without such degrees?**.

## 1.1 Source of the Data

Kaggle conducted their Annual Data Science survey and it was full of interesting questions. Participants of this survey were asked different questions about their demographics, profiles, companies, what they use etc. I analysed this data intending to dig deeper into the profiles of people who completed the university degrees to learn data science and those who did not. The focus of the study is to identify if the working data scientists with official higher education degrees differ significantly from the other group. The analysis and storyline are segmented according to different factors.

NOTE

The respondents who were "students" and "not employed" were removed for analysis. The two groups were selected based on the respondent's choice if they completed the university degrees to learn data science or not.

# 2

# Data Overview

To evaluate the return on investment (ROI) of pursuing a Master's degree in Data Science, this study utilizes a structured dataset containing detailed information about data professionals across the globe. The data includes demographic, educational, professional, and compensation-related variables, enabling a multi-faceted exploration of how various factors influence salary outcomes in the data science job market.

## 2.1 Dataset Structure

The dataset comprises several key columns, each representing an important aspect of the respondent's background or employment. The most relevant fields used in this analysis include:

- **Q5** – Country of residence
- **Q6** – Highest level of formal education completed
- **Q8** – Current employment status
- **Q15** – Job title or current role
- **Q24** – Industry of employer
- **Q25** – Size of the employing organization
- **Q10** – Years of experience in coding
- **Q11** – Years of experience in current role
- **Q29** – Annual compensation (converted to USD)

These columns were selected for their direct relevance to the analysis of salary trends and educational outcomes. Additional variables (e.g., programming languages used, tools and platforms familiar to respondents) were omitted from this report to maintain focus on the core research objective.

## 2.2 Initial Filtering and Cleanup

Given the self-reported nature of the dataset, several preprocessing steps were applied to improve data quality and ensure reliable comparisons:

1. **Employment Status Filter:** Only responses indicating full-time employment were retained (`Q8 = 'Employed full-time'`). This excludes students, freelancers, and part-time workers whose income patterns may not be comparable.
2. **Salary Range Filtering:** Responses with unreported or ambiguous salary data were removed. To mitigate the influence of outliers and likely reporting errors, a salary range filter was applied:
   - Minimum salary: $5,000
   - Maximum salary: $500,000

   Salaries outside this range were either considered unrealistic or unrepresentative for the purposes of this study.
3. **Missing Data Handling:** Entries missing critical values (e.g., country, education level, or job title) were excluded. This ensured consistent group-level comparisons in the visual analysis.
4. **Standardizing Education Levels:** Education levels were grouped into four standardized categories:
   - Bachelor's degree
   - Master's degree
   - Doctoral degree
   - Other (includes professional certifications, associate degrees, etc.)

   This grouping helped simplify visualizations and reduce noise in comparisons.
5. **Experience Normalization:** Years of experience were extracted and converted to numerical format to enable correlation analysis with salary.
6. **Country-Level Aggregation:** Countries with very few respondents were either excluded or grouped under "Other" to avoid skewing visual representations.

## 2.3 Post-Cleaning Dataset Profile

After the cleanup process, the refined dataset included approximately:

- **20,000+** full-time respondents
- **Over 60** countries
- **Dozens of distinct job roles** — with a focus on common titles such as Data Scientist, Data Analyst, Machine Learning Engineer, and Software Engineer

This cleaned and structured dataset forms the basis for the salary distribution analysis, role-specific comparisons, and education-based insights discussed in subsequent sections of the report.

# 3

# Sources of Learning Data Science

Data science skills are straightforward to obtain, they need experience and learning. Nowadays there are many online and offline which teaches them in detail. While some prefer online courses such as Coursera or Udacity, some prefer to go to universities for a year or two-year long dedicated courses. Let's look at what are the most popular sources of learning data science among the respondents of the kaggle survey. In this question, one participant could have chosen multiple choices, hence the x-axis represents "percentage" of respondents who selected a particular choice.

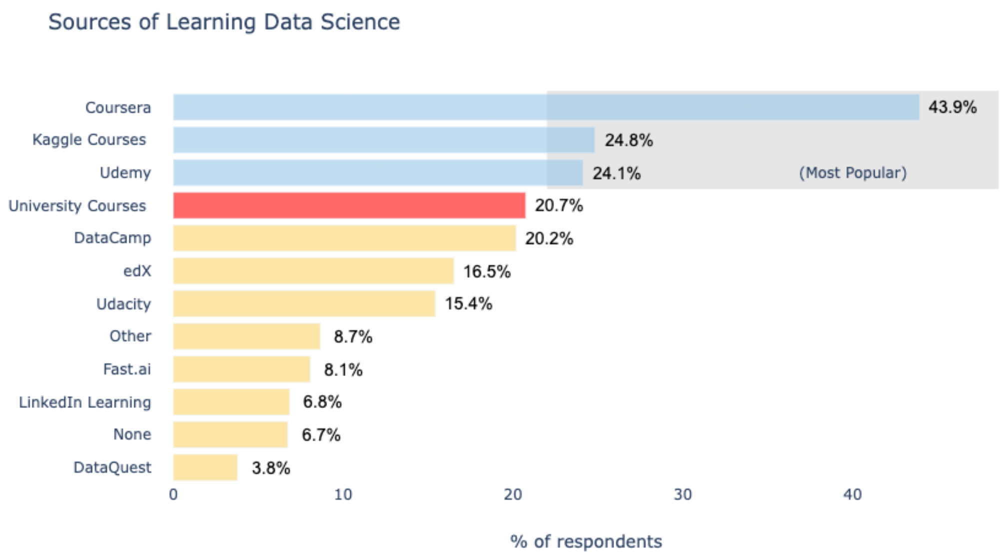



**Figure 3.1:** *Sources of learning.*

**Key Findings**

1. About 44% of the respondents selected ”Coursera” as the primary source of learning data science. Coursera is the popular online learning platform which provides both free courses and paid specializations.
2. Coursera and its founder Andrew NG have made very significant contributions to the data science revolution. Back in 2012, Andrew NG released the very popular Machine Learning course which became the first choice for many to learn data science. In 2017, Deeplearning.ai was launched and it became very popular data science specialization. These courses and many others from well known academic names on the online platform makes Coursera as the primary choice among the data science enthusiasts.
3. Kaggle Learn was selected by almost one-fourth of the respondents. Kaggle team launched these courses somewhere around early 2017. They are composed of notebook style materials which not only focusses on teaching the concepts but also the programming part as well.
4. Then there are other sources such as Udemy, Udacity, edX, fast.ai etc. These platforms als provide online materials and courses to learn data science.
5. Then there is a group of individuals who prefer to go to a university to pursue higher education degrees. Among the survey participants, about **one-fifth of the participants** had completed university degrees. Accoding to multiple sources ( Masters-And-More, CareerAdditct, MyBaggage ) different individuals have many different reasons to choose university courses over online courses. The most common are:

    - Personal Goal
    - Better Salary
    - Gain More Knowledge
    - Better Job Roles
    - Career Change

## 3.1 Designing My Survey

I created my own survey to know the student's choices this question and shared it in several groups associated with National University of Singapore. I managed to get about 120 responses from different people for this survey. Following is the response distribution of the Question.

Link of the Survey: Survey

Link of the Responses: Responses

**NOTE**

Names are removed to maintain the privacy of the individuals.

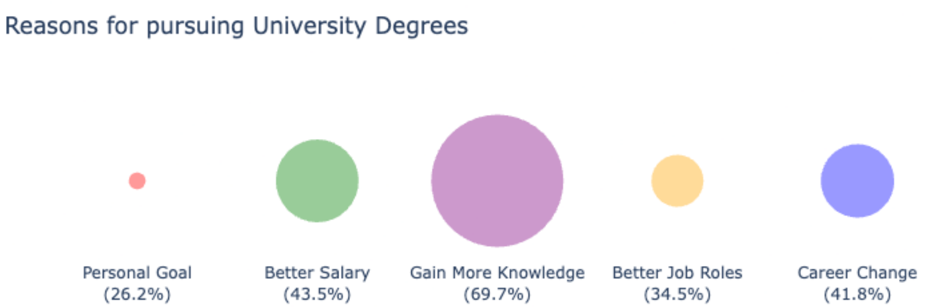



**Figure 3.2:** *Reasons for pursuing university degree.*

**Key Findings**

1. Most of the people decided to pursue higher education degrees to gain more knowledge. This was the primary reason for about 70% of the individuals who were part of this survey. Every two out of five people decided to pursue university degrees to get better salaries or to change their professions. Only about one-fourth of individuals had their own personal goal to go for a university degree.
2. If we just look at the number of individuals who selected "gain more knowledge", the same but important question arises again: "Is it worth spending a huge amount of money to gain knowledge that one can get from free sources?" Arent' the free online sources good enough to gain more knowledge. Or, Can't

these sources provide enough skills and knowledge to get that better salary, better job roles, or provide a pathway for a career change.

3. The interesting fact to note that most of the online courses on Coursera, Udemy etc. are also from the same universities or the same professors. Many of them are free as well. So if ”gaining more knowledge” is the only goal then it is worth considering these free courses.

## 3.2 The Academic Landscape

Whatever be the debate but one point is definitely clear, Universities across the globe do benefit a lot from this increased interest in higher education degrees. Many universities have now started specialized masters degree programs in analytics, data science, business analytics etc. The following chart shows some of the popular master's degree programmes from US universities along with their tuition fee and duration. Some of them are provided online, but the same are also provided on campus.

**The source of the data being:** https://www.kdnuggets.com/2019/04/best-masters-data-science-analytics-online.html

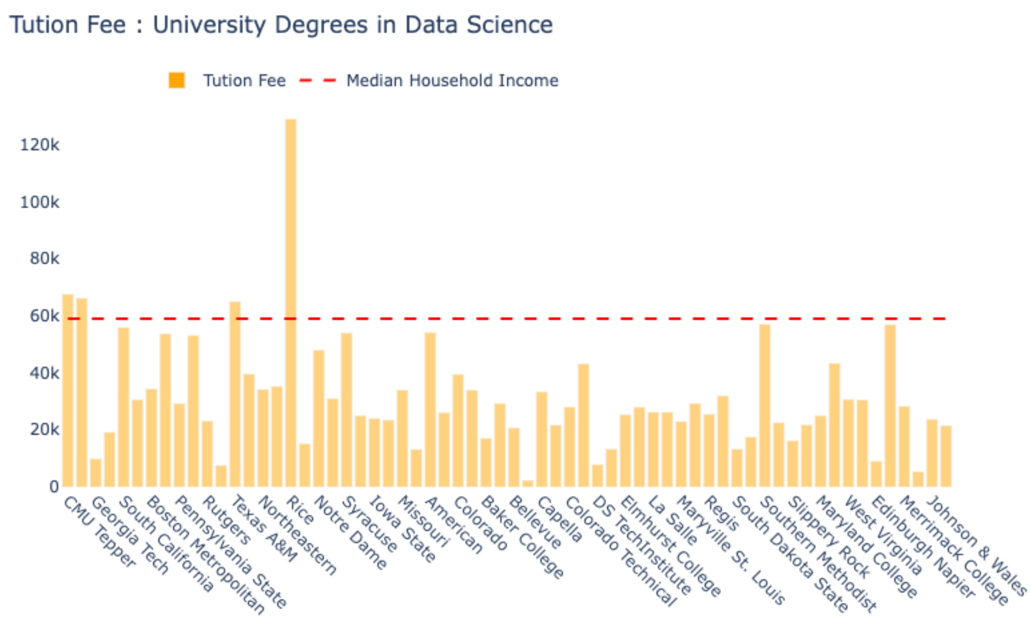



**Figure 3.3:** *Tuition-fees.*

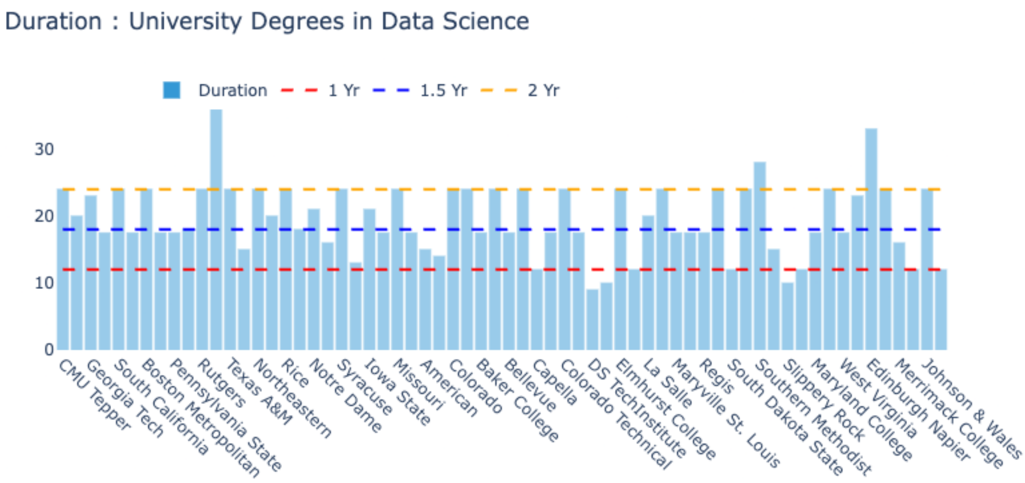



**Figure 3.4:** *Duration.*

Time and money are the two biggest investments associated with university degrees. The graph shows that the tuition fee for most of these courses is not cheap and the duration can range from anywhere 1 to 3 years depending upon specialization, location, and university type.

The university courses are of two types:

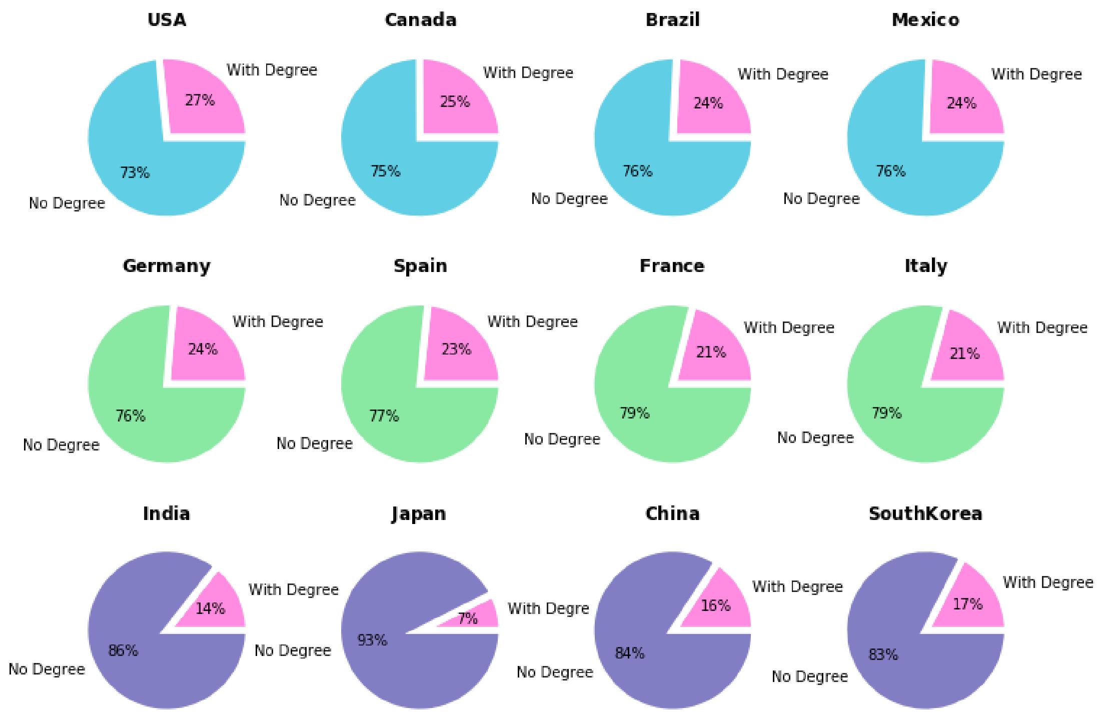
- **Generic courses** are typically very comprehensive, they cover all parts of data science but they are not very deep and detailed. These type of courses are good for those who want to get acquainted with main elements of this field.
- The **specializations**, on the other hand, aim to cover every possible detail of one particular area. They are generally very deep.

For both types of courses, the investment of money and time are always higher as compared to the alternative free ones.

> Head of Data Science from Restaurant Technologies, Inc. shared, ”No single Masters Program could cover all the disciplines needed in significant depth for one to be an expert in all these areas. Selecting an area or two or three and having depth and expertise in those is common. Many companies do not have just a ”Data Scientist” but teams comprised of experts from the different disciplines.”

## 3.3 Proportion of Individuals with University Degrees

Let's look at what per cent of individuals completed their university degrees to become a data scientist across different countries. Respondents were asked about their country in one of the questions.




**Key Findings**

1. In the American continent, the USA and Canada are the two most sought places to pursue higher education degrees. About one-fourth of the respondents from these countries have completed their university degrees to becoming a data scientist. In the United States, about 27% of the individuals who were part of this survey completed their university degrees. In Europe, there are about one-fifth of respondents who completed their university degrees while in Asia, the percentage is a bit lower only about 15%.
2. Countries with the highest proportion of data scientists with university degrees are **'Tunisia'**, **'Austria'**, **'New Zealand'** and **'Greece'** with **over 40% of the individuals completing university degrees**. On the other hand, countries **'Japan'**,

**'Nigeria'**, **'Belarus'**, and **'Algeria'** shows a lower number (less than 10%) of individuals completing university degrees.

3. Among the genders, female respondents have a higher number for completing university degrees than male respondents. **23% of the female respondents** and 20% of the male respondents completed their university degrees.

# 4

# Comparision between "With" and "Without University Degrees?"

According to Forbes, Most people with data science job titles don't have these new degrees. I also looked at the profiles of a few data scientist in my Linkedin Network and observed that not all of them have data science degrees. People tend to take different paths - some have degrees in business, economics, maths etc, while some have specialed data science degrees, and some have no university degree. But all of them are working in good organizations with good job roles. In the next section, let's look at the key insights from the Survey Data Analysis. The focus of the analysis is to compare the two groups - individuals who completed university degree vs those without for becoming a data scientist and identify key differences (if any).

Mainly, We will look at two perspectives: Are there a fairly equal percentage of individuals from two groups:

- With every compensation bracket.
- For each type of job role or activity.

## 4.1 Compensation

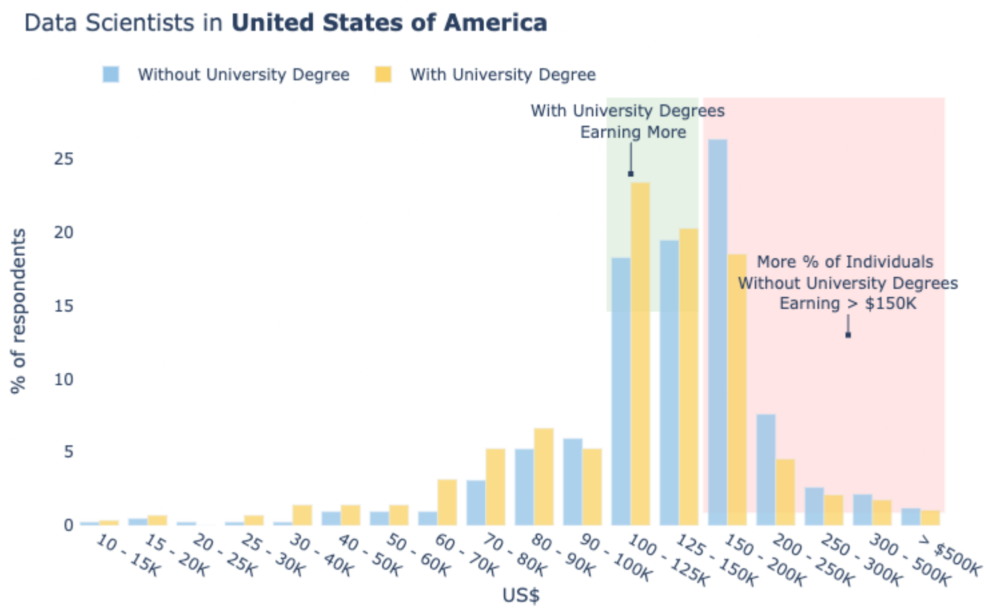



**Figure 4.1:** *Compensation*

### Key Findings

The plot shows the percentage of respondents from the United States of America in each compensation bracket. Looking at every bucket, it is clear that there are no significant differences between the two groups. Approximately they differ by a few per cent (less than 5). However, a few sections in this chart are very interesting.

> A common belief about university degrees is that one get higher compensation. The chart shows that there is a large percentage of individuals **without a university degree also earning more than 100K USD**. Even without university degrees, if individuals manage to obtain the right skills and the right direction, one can also grab high compensation. That's where Kaggle Learn or Coursera are the best options. As they provide the pathway to get the appropriate skills required to become a data scientist.

There are slight differences when the compensation is less than 125K. For this range, there is a higher percentage of individuals having university degrees. The area is shown in the green section in the chart. Well, this compensation range is the average of most of the companies in USA. This means that university degrees in data science can give an initial boost to the candidates in the compensation.

Very Interesting to note that there are more percentage of respondents without university degrees than those who have who are earning in the range of USD 150K-300K. This area is highlighted in red. This implies that there are definately ways to get higher compensation not necessarily after obtaining a masters in data science degree. The obvious reasons can be the amount of experience, age group, or special talent. It will be interesting to specifically look into this cohort where data scientists earn >150K USD. This is analysed in the next section.

### 4.1.1 Key Characteristics : Data Scientists earning > 150K

The following graph shows the key characteristics: age distribution, coding experience (in years) etc. for Data Scientist earning more than 150K USD.

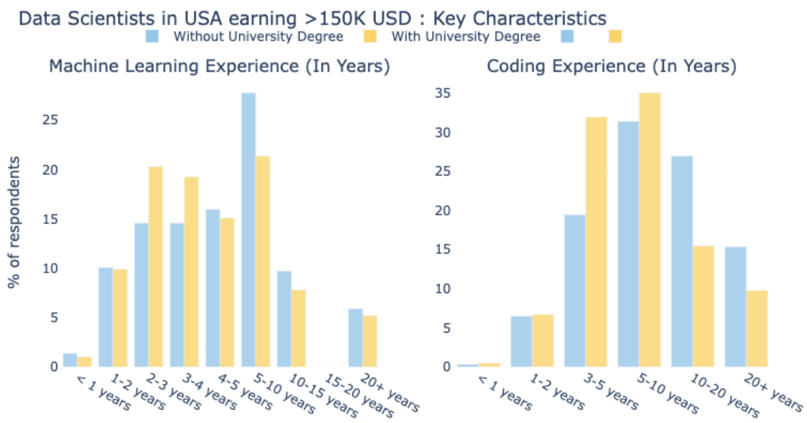



**Figure 4.2:** *Experience of data scientists earning more than 150K USD.*

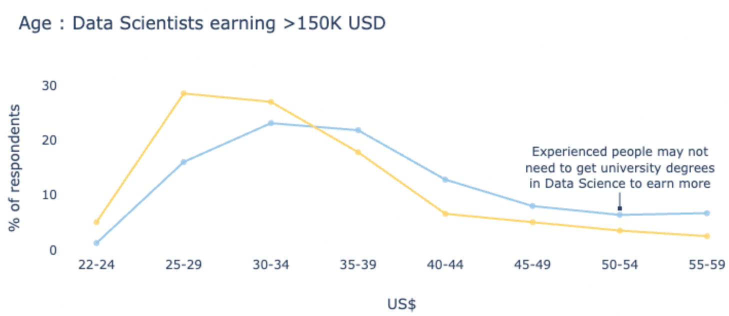



**Figure 4.3:** *Age data scientists earning more than 150K USD.*

**Key Findings**

1. The chart shows that there is a considerable number of individuals in each bracket of the machine learning experience, coding experience, and age. The charts are also does not shows any skewness in a particular bracket.
2. With more number of years in experience for coding and machine learning (greater than 5), We see that relatively a higher percentage of people are there earning more than 150K without university degrees.
3. More number of individuals who are aged less than 34 years and with a university degree earn greater than 150K. Good to see that even more percentage of young individuals, aged 22-24 also get higher compensation in the USA.
4. The portion on the right side of the age graph shows that experienced people may not need to get university degrees in data science if they are only looking for better salaries.
5. These trends are also similar in other countries. According to many links, top countries for higher education university degrees are the USA, Germany, Canadas etc.

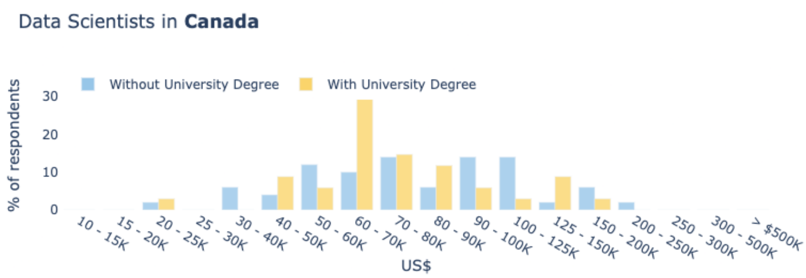



**Figure 4.4:** *Experience of data scientists earning more than 150K USD in Canada.*

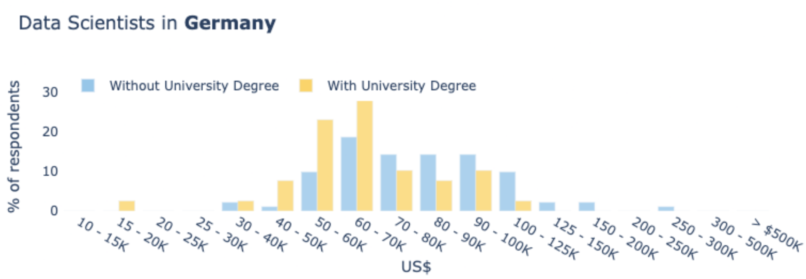



**Figure 4.5:** *Experience of data scientists earning more than 150K USD in Germany.*

We see similar insights for these countries as the USA where more percentage of non degree holder respondents are earning higher compensation in many brackets.

## 4.2 Job Role

Next we look at the job roles of individuals. Does it make a difference in terms of what kind of activity the data scientists do on the daily basis if they are coming with a university degree as compared to without.

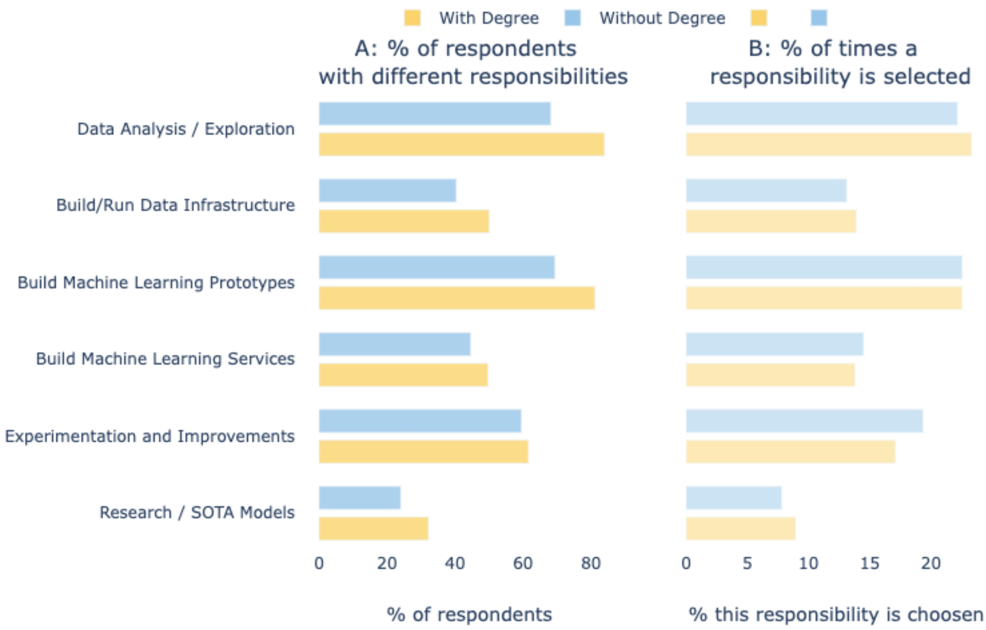



**Figure 4.6:** *Roles of data engineers.*

The two plots capture two different pieces of information:

**Plot A** shows what percentage of respondents selected a particular activity which people do in their day to day activities. This plot shows which are the most common activities people do in their data science project and the difference between a degree and non-degree holders.

**Plot B** shows, out of all the selected choices by all the respondents, what percentage is a particular responsibility is selected. This plot shows, for a particular group (degree or non-degree holders), which activity has more importance. For example, If a particular 'responsibility' is rarely selected, then the overall percentage will be smaller and If it is selected most of the times, then its overall percentage will also be higher. Meaning that it will be important to the particular group - degree or non-degree holders.

**Key Findings**

1. Plot A shows that a higher percentage of individuals with a university degree are involved in different tasks. The biggest differences are observed for data scientists who do "Data Analysis or Exploration", and performing "Research and developing State of the art models". Since experimentation is always a big part of any data science project, there exists very less differences in the % of respondents of two groups.
2. The percentage of non-degree holder respondents is always lesser than the counterpart, however, the differences are not extreme. There is still a significant percentage of people who are involved in similar responsibilities as the university holders. Hence, it will be wrong to say that only university holders do a particular type of activities or have specialized activities.
3. Plot B shows a higher percentage of people with university degrees selected being involved in **Research work, Data Analysis, and Building/Running Infrastructures**. While a slightly more percentage of people without university degrees selected being involved in Experimentation and Building Machine Learning services.
4. A fairly equal percentage is observed for people who build Machine Learning service despite their degrees. This particular selection shows that there is no major difference in the kind of work a data scientist will do.

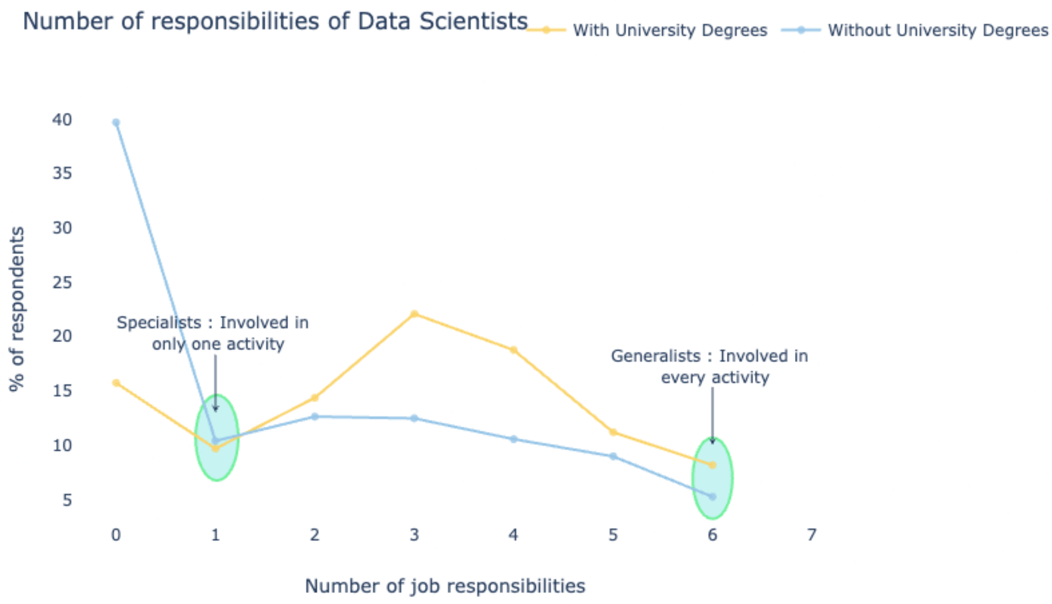



**Figure 4.7:** *responsibilities of a data scientist.*

### Key Findings

1. Based on the number of job responsibilities, a data scientist can be classified into two categories - Generalist and Specialist. Generalist are the individuals involved in all parts of life cycle of a data science project (ie. the points on the extereme right). Specialists are the individuals who are focussed on hardly 1 or 2 job responsibilities, ie. (points in the left).
2. We see that slightly more percentage of generalists are there having university degrees. They are they people who are involved in every step - doing analysis, experimentation, building services, and also research. In general, the plot shows that more percentage of people who completed university degrees are involved in multiple responsibilities.

# 5

# Identifying Key Traits

In this particular task, we aim to identify what are the most important signals strongly related to university degree holders. To identify these signals, we will treat this task as a predictive modelling problem. The first step is to prepare dataset which includes creating the train and test sets. Next, a simple learning classifier will be trained on the dataset the features are information about the individuals (demographics, tools used, company info etc.) and the target determines if the individual completed the university degree or not. All of the features and the target are binary. After the model is trained with a decent evaluation metric score, the important features of the model are obtained using Permutation Importance. These features provide some sense about their importance with an order or ranking.

```
X = one_hot_df[features]
y = one_hot_df["Q13_Part_10"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25,
↪  random_state = 2)
model = RandomForestClassifier(n_estimators=50, random_state=0).fit(X_train, y_train)
y_pred = model.predict(X_test)
print ("Classifier Accuracy: ", accuracy_score(y_pred, y_test))
```

**Listing 5.1:** *Identifying key traits of university degree holders.*

```
Classifier Accuracy:  0.7997832565700352
```

**Listing 5.2:** *Classifier Accurary*

The classifier shows a decent accuracy score of about 80%, let's now look at the permutation importance of every variable.

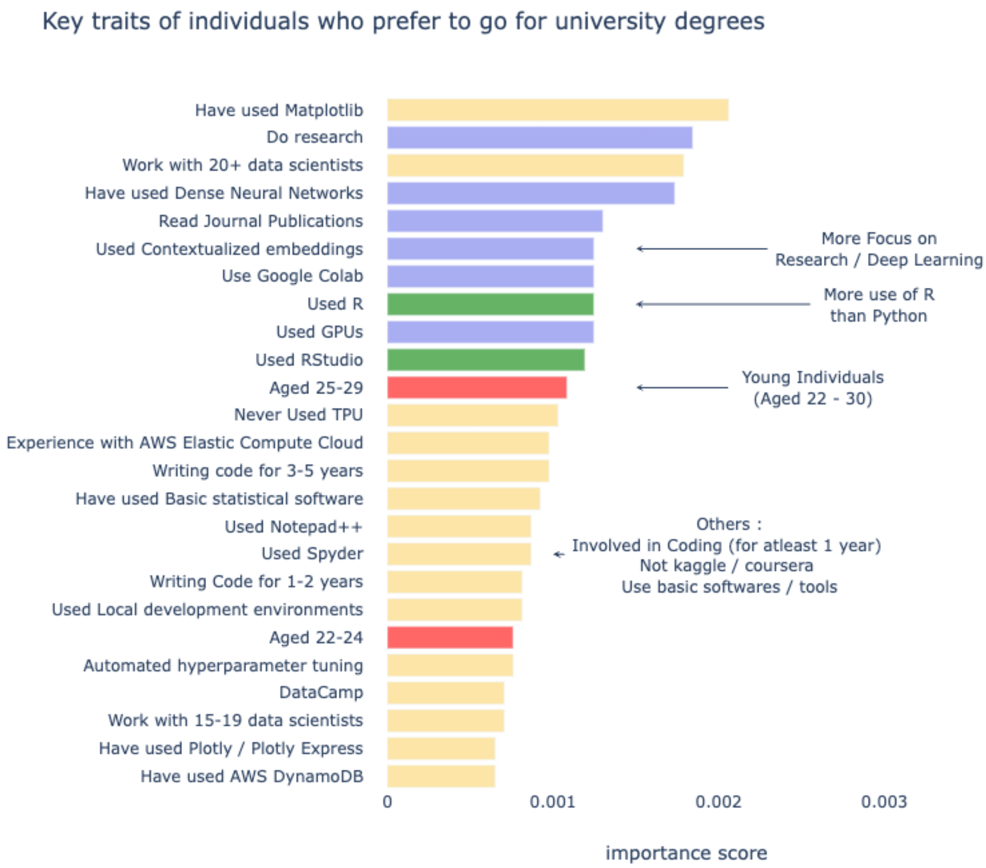



**Figure 5.1:** *Key traits of individuals who prefer to go for university degrees.*

The chart shows a few interesting traits of people associated with university degrees: More use of R than Python, More focus on research and deep learning, mainly young individuals (aged 22-30), involved in coding for at least a year, not doing kaggle or MOOCs.

# 6

# Conclusion

The main focus of this analysis was to understand and evaluate if there are any significant differences in the self-taught scientists and university degree holders. The insights presented in this notebook, convey a broad message about the importance of ”masters in data science degrees”. We can observe that it is not impossible to achieve certain job roles or position and even compensation without any degree in data science/analytics. In Section 4 we saw that there are as many as self-made data scientists (without any degree) earning similar compensation as with the degree holders. The case is not just restricted to the USA but also true in many other countries. This means that organizations value skills and talent over any degree, that's why they pay huge chunks of money for that skill. We also observed that there are no major distinctions in the type of work individuals in this field do on a daily basis, everyone has to spend a good amount of time in data analysis and cleaning, There are almost equal percentage of respondents who build machine learning services, prototypes and also do experimentation.

It might be wrong to strongly say that it's not worth it all. There are many positives associated with it as well: One can get edge during job hunt over those who do not have masters degree, one can establish their network with professors, fellow students, and industry partners, one can get exposure to many new areas, and can polish a lot of skills. Academic institutions are responding to a market demand and generating programs, due to it. It is possible that what students learn might be very different from what a data scientist does.

Jeremy Howard said the following in a reddit AMA:

> There is nothing that you could learn in a master of computer science that you could not learn in an online course, or by reading books. However, if you need the structure or motivation of a formal course, all the social interaction with other students, you may find that useful. I think that online courses are the best way to go if you can, because you can access the world's best teachers, and have a much more interactive learning environment — for example the use of short quizzes scattered throughout each lecture. Specifically, the courses that I would recommend our the machine learning course on Coursera, the introduction to data science course on Coursera, and the neural networks for machine learning course — surprise surprise, also on Coursera! Also, all of the introductory books here are a good choice. The most important thing, of course, is to practice download some open datasets from the web, and try to build models and develop insights, and post them on your blog. And be sure to compete in machine learning competitions — that's the only way to know whether your approaches are working or not.

The key takeaway is that one can become a data scientist (or any other similar position in this field) with right skills and can also manage to get decent compensation even without going through "Masters in Data Science (or similar degrees)". There are many alternatives - Coursera, Kaggle, Udemy, Udacity, eDX, Linkedin Learning, Fast.ai to name a few. These resources are either completely free or are much cheaper than dedicated university degrees. Additionally, one can complete these courses in a much shorter time as compared to the university courses. If there are any other reasons for the desire to pursuing masters, example - personal goals, passion, interest, utilizing the breaks etc, one should consider pursuing university courses.

If someone is thinking of getting a degree, in data science, analytics, just because they have the time and money to spare, is not wise. If the interest in analytics is driven by the desire for a job that offers better rewards than what they are getting now, then one should look for alternative paths to make that career move. Meta S Brown from Forbes says, "At the end, we need to remember that Data Scientist is a Title. Many give themselves or have this title because that's the work they do, not because they have a particular degree".

It’s just about taking the first step, start studying from online courses, do relevant projects, start making kaggle contributions, and keep improving your resume and profile. In the end, whatever path someone takes, they will most likely produce similar outcomes. The following Sankey depicts this message very well.

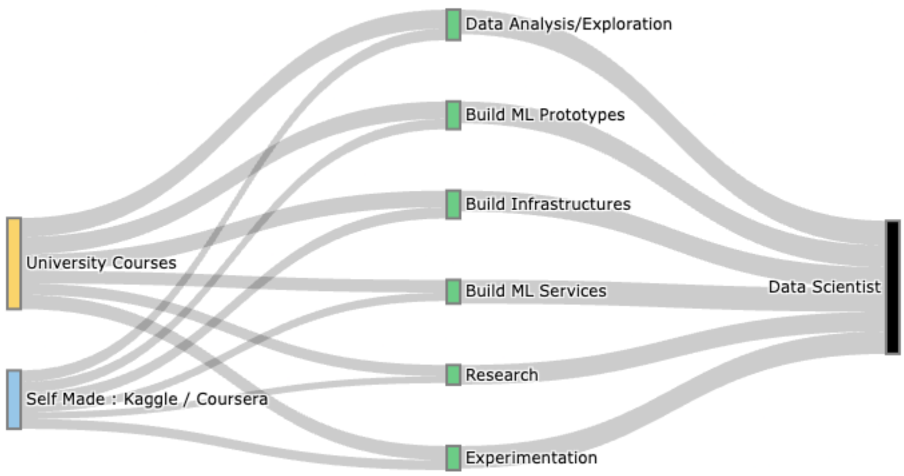



**Figure 6.1:** *Sankey diagram showing the distribution of data scientists with and without university degrees.*

## Summary

- If someone wants to learn data science skills only, then no its not worth it because there are many excellent free / cheaper alternatives on the internet. Kaggle is an excellent place to start.
- If someone wants to work as a data scientist (fresher or career change), then it may or may not be worth it because there is no gurantee that one will get a data science job after the university degree. A non degree holder with right skillset, right experience, and right attitude may get a better job than the one with a degree title.
- If someone wants to earn more, then it is not necessary to have university degree, as one with good experience or good skills can grab very high compensation

packages.

- If someone wants to do more research, then it may be worth it given that one chooses a right specialization.
- If someone is passionate to learn more and gain exposure then it is definately worth it. University degree can provide a platform to network, participate in different events, and improve the foundation.

# Appendices

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# A

# Appendix A

All the codes and datasets used in the text can be found in my GitHub repository named worth-of-Masters-in-DS.